SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Big Data Analytics: Hadoop Ecosystem, MapReduce & Apache Spark

Department: CSE DS	**Course / Subject:** Big Data Analytics (DS601PC)
Semester: Semester 6	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: 5 V's of Big Data, Hadoop Distributed File System (HDFS), MapReduce lifecycle, Apache Spark RDD transformations, Spark SQL DataFrames, and Streaming analytics.

Unit 1: Big Data Landscape & HDFS

Characteristics of Big Data (Volume, Velocity, Variety, Veracity, Value), HDFS architecture (NameNode, DataNode, Secondary NameNode), Block replication strategy.

Unit 2: MapReduce Programming Model

Map phase, Shuffle and Sort phase, Reduce phase, Combiner functions, Partitioner, Inverted index construction, and MapReduce word count tutorial.

Unit 3: YARN Architecture & Ecosystem

YARN Resource Manager, Node Manager, Application Master, Apache Hive data warehousing (HQL, Partitions, Buckets), and Apache Pig Latin.

Unit 4: Apache Spark Core Architecture

Limitations of MapReduce, In-memory computing, Resilient Distributed Datasets (RDD), Transformations vs Actions, Lineage graph, and Lazy evaluation.

Unit 5: Spark SQL & Machine Learning (MLlib)

Spark DataFrames, Spark SQL queries, Catalyst optimizer, Spark MLlib pipelines (Classification, Regression, Clustering), and Kafka real-time streaming ingestion.